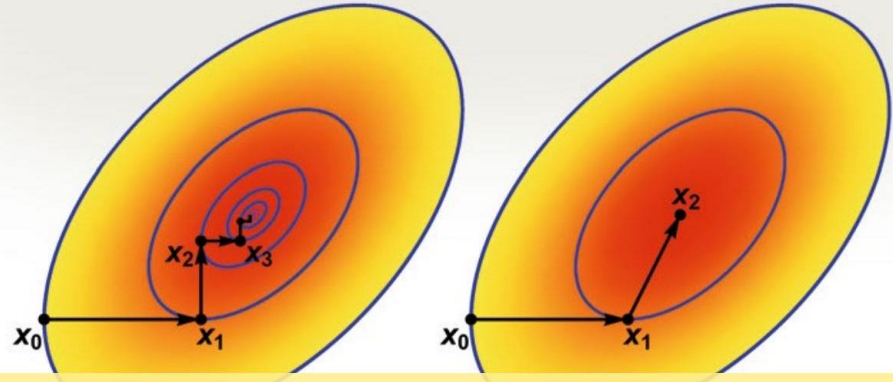


Image regions

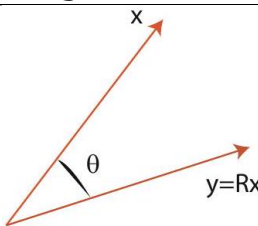
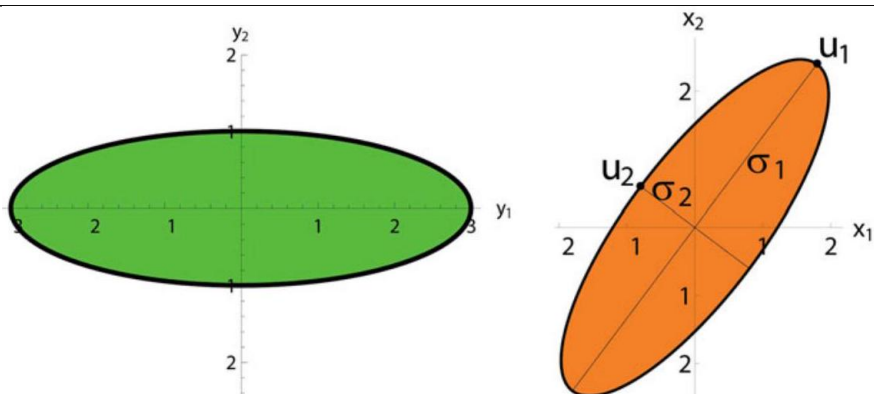
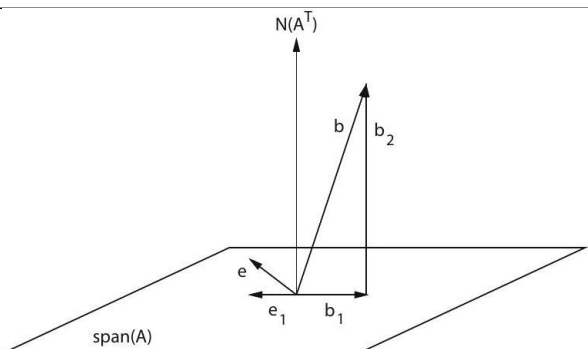
Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ DIA_0001	1	—	—	—	
lyche-numerical-linear-algebra_ DIA_0002	55	—	—	—	<pre>function [l,u]=trifactor(a,d,c) % [l,u]=trifactor(a,d,c) u=d; l=a; for k =1:length(a) l(k)=a(k)/u(k); u(k+1)=d(k+1)-l(k)*c(k); end</pre> Listing 2.1 trifactor
lyche-numerical-linear-algebra_ DIA_0003	55	—	—	—	<pre>function x = trisolve (l,u,c,b) % x = trisolve (l,u,c,b) x=b; n= size(b,1); for k =2:n x(k,:)=b(k,:)-l(k-1)*x(k-1,:); end x(n,:)=x(n,:)/u(n); for k=(n-1):-1:1 x(k,:)=(x(k,:)-c(k)*x(k+1,:))/u(k); end</pre> Listing 2.2 trisolve

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ DIA_0004	69	—	—	—	<pre> function [x,C]=splineint (a,b,y,mu1,munp1) % [x,C]=splineint (a,b,y,mu1,munp1) y=y(:); n=length(y)-1; h=(b-a)/n; x=a:h:b-h; c=ones (n-2,1); [l,u]= trifactor (c,4*ones (n-1,1),c); b1=6/h^2*(y(3:n+1)-2*y(2:n)+y(1:n-1)); b1(1)=b1(1)-mu1; b1(n-1)=b1(n-1)-munp1; mu= [mu1;trisolve (l,u,c,b1);munp1]; C=zeros (4*n,1); C(1:4:4*n-3)=y(1:n); C(2:4:4*n-2)=(y(2:n+1)-y(1:n))/h... -h*mu(1:n)/3-h*mu(2:n+1)/6; C(3:4:4*n-1)=mu(1:n)/2; C(4:4:4*n)=(mu(2:n+1)-mu(1:n))/(6*h); C=reshape (C,4,n)'; end </pre> Listing 2.3 splineint
lyche-numerical-linear-algebra_ DIA_0005	69	—	—	—	<pre> function i = findsubintervals (t,x) %i = findsubintervals (t,x) k=length(t); m=length(x); if k<2 i=ones (m,1); else t(1)=min (x(1),t(1))-1; [~,j]=sort ([t(:)',x(:)']); i=(find (j>k)-(1:m))'; end </pre> Listing 2.4 findsubintervals
lyche-numerical-linear-algebra_ DIA_0006	70	—	—	—	<pre> function [X,G]=splineval (x,C,X) % [X,G]=splineval (x,C,X) m=length(X); i=findsubintervals (x,X); G=zeros (m,1); for j=1:m k=i(j); t=X(j)-x(k); G(j)=[1,t,t^2,t^3]*C(k,:)'; end </pre> Listing 2.5 splineval
lyche-numerical-linear-algebra_ DIA_0007	77	—	—	—	The diagrams show the construction of the matrix $A^{(n)}$ for the spline interpolation. $A^{(1)}$ is a 1x1 red square. $A^{(2)}$ is a 2x2 matrix with a red square and a yellow rectangle. $A^{(k)}$ is a kxk matrix with a red square, a yellow rectangle, and a blue rectangle. $A^{(n)}$ is an nxn matrix with a red square, a yellow rectangle, and a blue rectangle.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-DIA_0008	81	—	—	—	<pre> function x=rforwardsolve(A,b,d) % x=rforwardsolve(A,b,d) n=length(b); x=b; x(1)=b(1)/A(1,1); for k=2:n lk=max(1,k-d); x(k)=(b(k)-A(k,lk:(k-1))*x(lk:(k-1)))/A(k,k); end </pre> Listing 3.1 rforwardsolve
lyche-numerical-linear-algebra-DIA_0009	81	—	—	—	<pre> function x=rbacksolve(A,b,d) % x=rbacksolve(A,b,d) n=length(b); x=b; x(n)=b(n)/A(n,n); for k=n-1:-1:1 uk=min(n,k+d); x(k)=(b(k)-A(k,(k+1):uk)*x((k+1):uk))/A(k,k); end </pre> Listing 3.2 rbacksolve
lyche-numerical-linear-algebra-DIA_0010	82	—	—	—	<pre> function x=cforwardsolve(A,b,d) %x=cforwardsolve(A,b,d) x=b; n=length(b); for k=1:n-1 x(k)=b(k)/A(k,k); uk=min(n,k+d); b((k+1):uk)=b((k+1):uk)-A((k+1):uk,k)*x(k); end x(n)=b(n)/A(n,n); end </pre> Listing 3.3 cforwardsolve
lyche-numerical-linear-algebra-DIA_0011	91	—	—	—	<pre> function [L,U]=L1U(A,d) % [L,U]=L1U(A,d) n=length(A); L=eye(n,n); U=zeros(n,n);U(1,1)=A(1,1); for k=2:n km=max(1,k-d); L(k,km:(k-1))=rforwardsolve(U(km:(k-1) ... ,km:(k-1))',A(k,km:(k-1))',d)'; U(km:k,k)=rforwardsolve(L(km:k,km:k),A(km:k,k),d); end </pre> Listing 3.4 L1U
lyche-numerical-linear-algebra-DIA_0012	93	—	—	—	<pre> function x=cbacksolve(A,b,d) % x=cbacksolve(A,b,d) x=b; n=length(b); for k=n:-1:2 x(k)=b(k)/A(k,k); lk=max(1,k-d); b(lk:(k-1))=b(lk:(k-1))-A(lk:(k-1),k)*x(k); end x(1)=b(1)/A(1,1); end </pre> Listing 3.5 cbacksolve

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-DIA_0013	102	—	—	—	<pre> function [L,dg]=LDLs(A,d) % [L,dg]=LDLs(A,d) n=length(A); L=eye(n,n); dg=zeros(n,1); dg(1)=A(1,1); for k=2:n m=rforwardsolve(L(1:k-1,1:k-1),A(1:k-1,k),d); L(k,1:k-1)=m./dg(1:k-1); dg(k)=A(k,k)-L(k,1:k-1)*m; end </pre> Listing 4.1 LDLs
lyche-numerical-linear-algebra-DIA_0014	106	—	—	—	<pre> function L=bandcholesky(A,d) %L=bandcholesky(A,d) [L,dg]=LDL(A,d); L=L*diag(sqrt(dg)); end </pre> Listing 4.2 bandcholesky
lyche-numerical-linear-algebra-DIA_0015	111	—	—	—	<pre> function L=bandsemicholeskyL(A,d) %L=bandsemicholeskyL(A,d) n=length(A); for k=1:n kp=min(n,k+d); if A(k,k)>0 A(k,k)=sqrt(A(k,k)); A((k+1):kp,k)=A((k+1):kp,k)/A(k,k); for j=k+1:kp A(j:kp,j)=A(j:kp,j)-A(j,k)*A(j:kp,k); end else A(k:kp,k)=zeros(kp-k+1,1); end end L=tril(A); end </pre> Listing 4.3 bandsemicholeskyL
lyche-numerical-linear-algebra-DIA_0016	120	—	—	—	

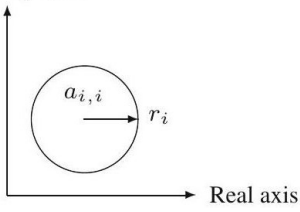
Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra DIA_0017	122	—	—	—	
lyche-numerical-linear-algebra DIA_0018	125	—	—	—	
lyche-numerical-linear-algebra DIA_0019	127	—	—	—	<pre> function [u,a]=housegen(x) % [u,a]=housegen(x) a=norm(x); if a==0 u=x; u(1)=sqrt(2); return; end if x(1)== 0 r=1; else r=x(1)/abs(x(1)); end u=conj(r)*x/a; u(1)=u(1)+1; u=u/sqrt(u(1)); a=-r*a; end </pre> Listing 5.1 housegen
lyche-numerical-linear-algebra DIA_0020	129	—	—	—	<pre> function [R,C] = housetriang(A,B) % [R,C] = housetriang(A,B) [m,n]=size(A); r=size(B,2); A=[A,B]; for k=1:min(n,m-1) [v,A(k,k)]=housegen(A(k:m,k)); C=A(k:m,k+1:n+r); A(k:m,k+1:n+r)=C-v*(v'*C); end R=triu(A(:,1:n)); C=A(:,n+1:n+r); end </pre> Listing 5.2 housetriang

Identifier	Page	Conf.	LaTeX source	Rendered	Image																											
lyche-numerical-linear-algebra- DIA_0021	134	—	—	—																												
lyche-numerical-linear-algebra- DIA_0022	140	—	—	—	<pre>function x=rothesstri(A,b) % x=rothesstri(A,b) n=length(A); A=[A b]; for k=1:n-1 r=norm([A(k,k),A(k+1,k)]); if r>0 c=A(k,k)/r; s=A(k+1,k)/r; A([k k+1],k+1:n+1) ... =[c s;-s c]*A([k k+1],k+1:n+1); end A(k,k)=r; A(k+1,k)=0; end x=rbacksolve(A(:,1:n),A(:,n+1),n); end</pre> Listing 5.3 rothesstri																											
lyche-numerical-linear-algebra- DIA_0023	175	—	—	—																												
lyche-numerical-linear-algebra- DIA_0024	225	—	—	—																												
lyche-numerical-linear-algebra- DIA_0025	240	—	—	—	<table><tr><td>1,1</td><td>1,2</td><td>1,3</td></tr><tr><td>2,1</td><td>2,2</td><td>2,3</td></tr><tr><td>3,1</td><td>3,2</td><td>3,3</td></tr></table> $v_{j,k}$ in V - matrix <table><tr><td>1,3</td><td>2,3</td><td>3,3</td></tr><tr><td>1,2</td><td>2,2</td><td>3,2</td></tr><tr><td>1,1</td><td>2,1</td><td>3,1</td></tr></table> $v_{j,k}$ in grid <table><tr><td>7</td><td>8</td><td>9</td></tr><tr><td>4</td><td>5</td><td>6</td></tr><tr><td>1</td><td>2</td><td>3</td></tr></table> x_i in grid	1,1	1,2	1,3	2,1	2,2	2,3	3,1	3,2	3,3	1,3	2,3	3,3	1,2	2,2	3,2	1,1	2,1	3,1	7	8	9	4	5	6	1	2	3
1,1	1,2	1,3																														
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1,3	2,3	3,3																														
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1,1	2,1	3,1																														
7	8	9																														
4	5	6																														
1	2	3																														

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra DIA_0026	240	—	—	—	
lyche-numerical-linear-algebra DIA_0027	241	—	—	—	
lyche-numerical-linear-algebra DIA_0028	254	—	—	—	<pre> function V=fastpoisson(F) %function V=fastpoisson(F) m=length(F); h=1/(m+1); hv=pi*h*(1:m)'; sigma=sin(hv/2).^2; S=sin(hv*(1:m)); G=S*F*S; X=h^4*G./(sigma*ones(1,m)+ ones(m,1)*sigma'); V=zeros(m+2,m+2); V(2:m+1,2:m+1)=S*X*S; end </pre> Listing 11.1 fastpoisson
lyche-numerical-linear-algebra DIA_0029	259	—	—	—	<pre> function z=ffttrec(y) %function z=ffttrec(y) y=y(:); n=length(y); if n==1 z=y; else q1=ffttrec(y(1:2:n-1)); q2=exp(-2*pi*1i/n).^(0:n/2-1)'.*ffttrec(y(2:2:n)); z=[q1+q2; q1-q2]; end </pre> Listing 11.2 fftrec
lyche-numerical-linear-algebra DIA_0030	262	—	—	—	<pre> function U=simplefastbiharmonic(F) </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-2019-DIA_0031	268	—	—	—	<pre> function k=jdp(F,K,tol) % k=jdp(F,K,tol) m=length(F); U=fastpoisson(F); V=zeros(m+2,m+2); E=F/(m+1)^2; for k=1:K V(2:m+1,2:m+1)=(V(1:m,2:m+1)+V(3:m+2,2:m+1)... +V(2:m+1,1:m)+V(2:m+1,3:m+2)+E)/4; if max(max(abs(V-U)))<tol, return end end k=K+1; end </pre> Listing 12.1 jdp
lyche-numerical-linear-algebra-2019-DIA_0032	269	—	—	—	<pre> function k=sordp(F,K,w,tol) % k=sordp(F,K,w,tol) m=length(F); U=fastpoisson(F); V=zeros(m+2,m+2); E=F/(m+1)^2; for k=1:K for j=2:m+1 for i=2:m+1 V(i,j)=w*(V(i-1,j)+V(i+1,j)+V(i,j-1)... +V(i,j+1)+E(i-1,j-1))/4+(1-w)*V(i,j); end end if max(max(abs(V-U)))<tol, return end end k=K+1; end </pre> Listing 12.2 sordp
lyche-numerical-linear-algebra-2019-DIA_0033	274	—	—	—	
lyche-numerical-linear-algebra-2019-DIA_0034	292	—	—	—	

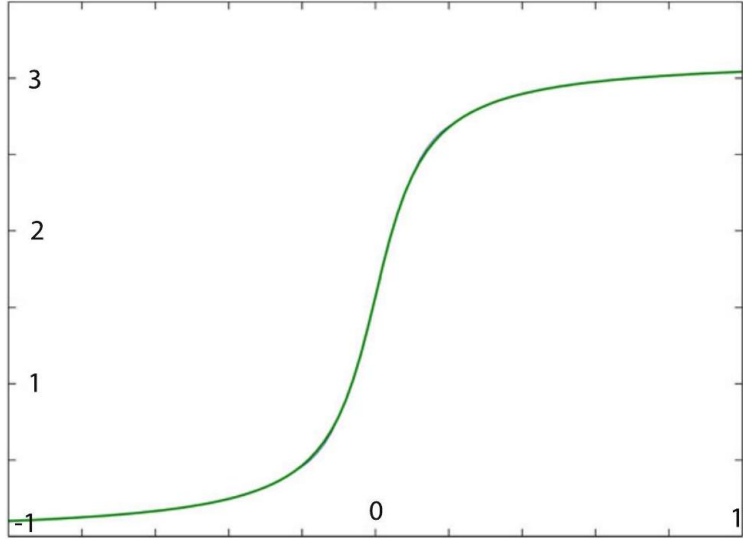
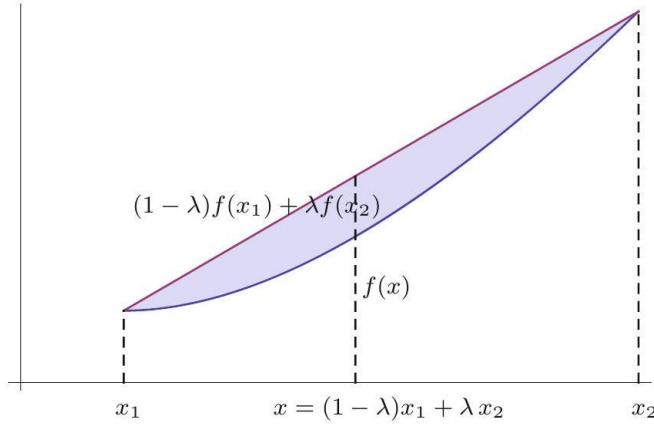
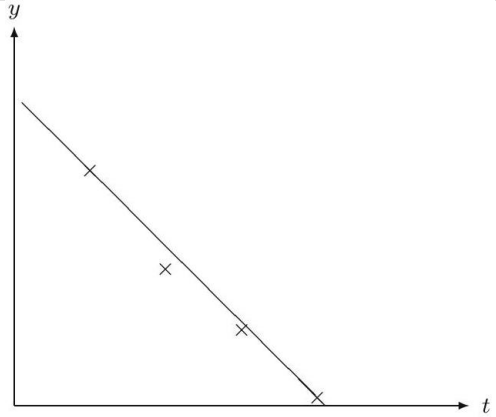
Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-DIA_0035	297	—	—	—	<pre> function [x,K]=cg(A,b,x,tol,itmax) % [x,K]=cg(A,b,x,tol,itmax) r=b-A*x; p=r; rho0=b'*b; rho=r'*r; for k=0:itmax if sqrt(rho/rho0)<= tol K=k; return end t=A*p; a=rho/(p'*t); x=x+a*p; r=r-a*t; rhos=rho; rho=r'*r; p=r+(rho/rhos)*p; end K=itmax+1; end </pre> Listing 13.1 cg
lyche-numerical-linear-algebra-DIA_0036	298	—	—	—	<pre> function [V,K]=cgtest(m,a,d,tol,itmax) % [V,K]=cgtest(m,a,d,tol,itmax) R=ones(m)/(m+1)^2; rho=sum(sum(R.*R)); rho0=rho; P=R; V=zeros(m,m); Tl=sparse(tridiagonal(a,d,a,m)); for k=1:itmax if sqrt(rho/rho0)<= tol K=k; return end T=Tl*P+P*Tl; a=rho/sum(sum(P.*T)); V=V+a*P; R=R-a*T; rhos=rho; rho=sum(sum(R.*R)); P=R+(rho/rhos)*P; end K=itmax+1; end </pre> Listing 13.2 cgtest
lyche-numerical-linear-algebra-DIA_0037	302	—	—	—	

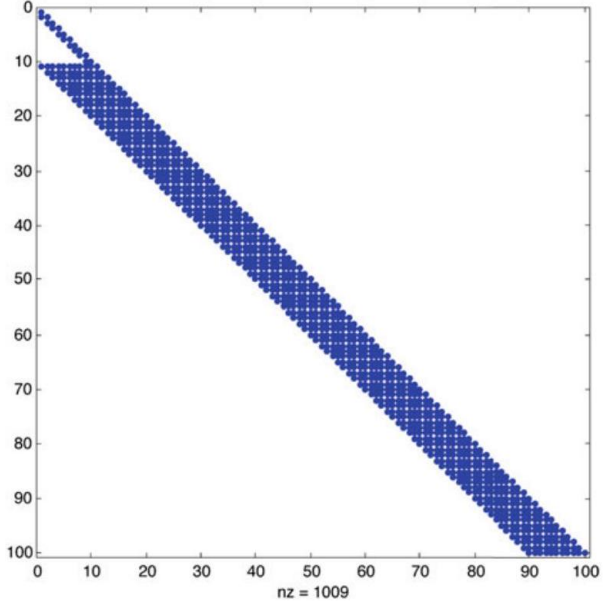
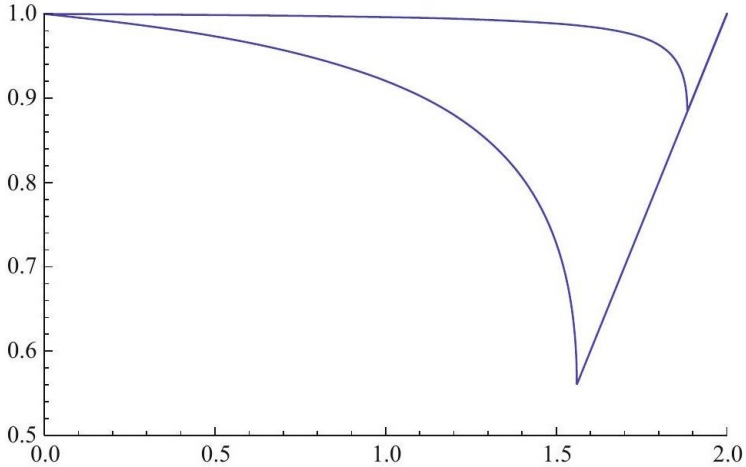
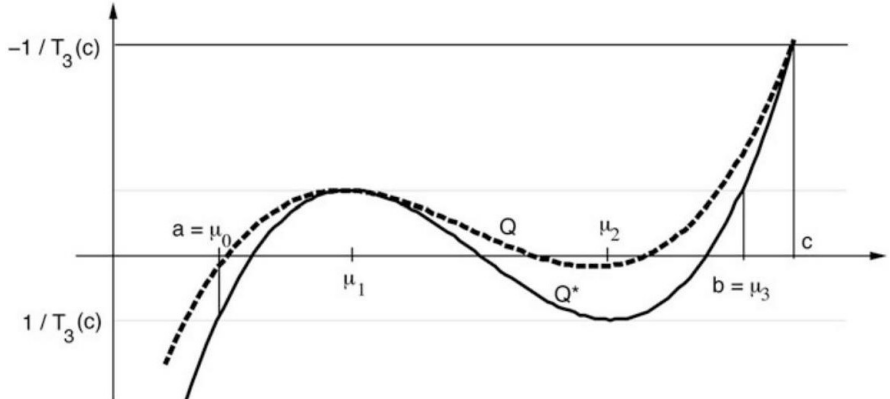
Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra DIA_0038	312	—	—	—	<pre> function [x,K]=pcg(A,B,b,x,tol,itmax) % [x,K]=pcg(A,B,b,x,tol,itmax) r=b-A*x; p=B*r; s=p; rho=s'*r; rho0=b'*b; for k=0:itmax if sqrt(rho/rho0)<= tol K=k; return end t=A*p; a=rho/(p'*t); x=x+a*p; r=r-a*t; w=B*t; s=s-a*w; rhos=rho; rho=s'*r; p=r+(rho/rhos)*p; end K=itmax+1; end </pre> Listing 13.3 pcg
lyche-numerical-linear-algebra DIA_0039	328	—	—	—	Imaginary axis  Real axis
lyche-numerical-linear-algebra DIA_0040	334	—	—	—	<pre> function [L,B] = hesshousegen(A) n=length(A); L=zeros(n,n); B=A; for k=1:n-2 [v,B(k+1,k)]=housegen(B(k+1:n,k)); L(k+1:n,k)=v; B(k+2:n,k)=zeros(n-k-1,1); C=B(k+1:n,(k+1):n); B(k+1:n,(k+1):n)=C-v*(v'*C); C=B(1:n,(k+1):n); B(1:n,(k+1):n)=C-(C*v)*v'; end end </pre> Listing 14.1 hesshousegen
lyche-numerical-linear-algebra DIA_0041	335	—	—	—	<pre> function Q = accumulateQ(L) n=length(L); Q=eye(n); for k=n-2:-1:1 v=L(k+1:n,k); C=Q(k+1:n,(k+1):n); Q(k+1:n,(k+1):n)=C-v*(v'*C); end </pre> Listing 14.2 accumulateQ

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra DIA_0042	339	—	—	—	<pre> h = b - a; for j = 1 : itmax c = (a + b)/2; if b - a < eps * h λ = (a + b)/2; return end k = ρ(c); if k ≥ m a = c else b = c; end </pre>
lyche-numerical-linear-algebra DIA_0043	347	—	—	—	<pre> function [l,x,it]=powerit(A,z,K,tol) % [l,x,it]=powerit(A,z,K,tol) af=norm(A,'fro'); x=z/norm(z); for k=1:K y=A*x; l=x'*y; if norm(y-l*x)/af<tol it=k; x=y/norm(y); return end x=y/norm(y); end it=K+1; end </pre> Listing 15.1 powerit
lyche-numerical-linear-algebra DIA_0044	350	—	—	—	<pre> function [x,s,nr]=rayleighit(A,x,s) % [x,s,nr]=rayleighit(A,x,s) n=length(x); y=(A-s*eye(n,n))\x; yn=norm(y); w=x/yn; x=y/yn; rho=x'*w; s=s+rho; nr=norm(w-rho*x); end </pre> Listing 15.2 rayleighit
lyche-numerical-linear-algebra DIA_0045	354	—	—	—	$A_1 = A$ for $k = 1, 2, \dots$ Choose a shift s_k $Q_k R_k = A_k - s_k I$ (QR factorization of $A_k - s_k I$) $A_{k+1} = R_k Q_k + s_k I.$ end

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_PIC_0001	43	—	—	—	
lyche-numerical-linear-algebra_PIC_0002	48	—	—	—	
lyche-numerical-linear-algebra_PIC_0003	48	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_ PIC_0004	50	—	—	—	
lyche-numerical-linear-algebra_ PIC_0005	53	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_PIC_0006	70	—	—	—	
lyche-numerical-linear-algebra_PIC_0007	199	—	—	—	
lyche-numerical-linear-algebra_PIC_0008	215	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- PIC_0009	251	—	—	—	
lyche-numerical-linear-algebra- PIC_0010	278	—	—	—	
lyche-numerical-linear-algebra- PIC_0011	306	—	—	—	
lyche-numerical-linear-algebra- PIC_0012	339	—	—	—	$\lambda = (a + b)/2$; return
lyche-numerical-linear-algebra- PIC_0013	354	—	—	—	$\mathbf{A}_{k+1} = \mathbf{R}_k \mathbf{Q}_k + s_k \mathbf{I}.$